

Name: I Forgot Again Team Date: 4/5/24 Class: Engineering

MyDesign Portfolio

Element E: Application of STEM Principles and Practices

STEM Principles and Practices to Design Requirements (E1)

STEM Principle for High-Priority Design Requirement #1

STEM Concept: **Geometry**

Describe how STEM concept applies directly to the specifics of your design:

Geometry features a wide range of measurements like angles and the imperial system to help accurately define the properties of an object. We require the formula the volume of a rectangular prism (length x width x height) and the area of the base (length x width for the measurements).

STEM Principle for High-Priority Design Requirement #2 FOR DESIGN REQUIREMENT #3

STEM Concept: **Engineering**

Describe how STEM concept applies directly to the specifics of your design:

Engineering, the branch of science concerned with the designing and building of ideas to help solve problems, is a very broad STEM principle, as almost everything that surrounds us can be characterized as engineering, hence the term “engineering is everywhere.” This connects to our project because we must use design principles like iteration and prototyping to create our design to keep drivers safe.

STEM Principle for High-Priority Design Requirement #3 FOR DESIGN REQUIREMENT #2

STEM Concept: **Physics**

Describe how STEM concept applies directly to the specifics of your design:

Physics is the field of study that involves movement and the properties of matter like gravity. Physics is key to understanding our world and our design because the properties used in our design use physics. We will use the acceleration formula (acceleration = Change in velocity/time) to calculate the speed of the car to see if our items are at risk, and the formula for gravity (Force = Gravity constant x Mass of object 1 x mass of object 2 / distance²) to calculate the damage of dropping your items.

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Proposed Solution to STEM Principles and Practices (E2)

STEM Principle for Design Solution #1

STEM Concept: **Geometry**

Describe how STEM concept applies directly to the specifics of your solution:

In our design, we will properly apply accurate measurements and angles to our prototype to optimize our solution. Any crooked sides or disproportionate measurements could result in an unsuccessful design.

STEM Principle for Design Solution #2

STEM Concept: **Engineering**

Describe how STEM concept applies directly to the specifics of your solution:

Engineering is known to be “everywhere”, and is unsurprisingly present in our design. To conclude with our final prototype, we brainstormed, incorporating math and science principles such as geometry and physics. We also considered our design requirements to narrow our options to select our best design.

STEM Principle for Design Solution #3

STEM Concept: **Physics**

Describe how STEM concept applies directly to the specifics of your solution:

While adding to our initial design, we realized that the objects would continue to move when the vehicle made turns or simply when it passed over speedbumps. Thus, we added foam padding to hold all items in place better and reduce the risk of damaging the items.

Evidence Design is Reviewed (E3)

Documentation #1: Sid

- Face to Face conversation
- Said how mathematics and physics can be applied to our engineering and our design. He also provided feedback on how to add safety features like a door/flap to our design.
- Sid’s feedback told us that we should use our mathematical measurements to have different sizes of compartments in our designs. We can also use physics to test the security of our items and use engineering to encompass our design.

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Documentation #2: Mr. Stienhorn

- Face to Face conversation
- Explained how we would need an equation for measurements and for the car and how we should think about how the driver uses the device. For example, to have the correct measurements in our device, we must think about the driver's needs.
- Mr. Stienhorn recommended that we use design principles in construction and equations in defining the problem to have the best design. We should also think about how these values affect the driver and the device.

Experts with Credentials (E4)

Expert #1:

Sid has an engineering degree and his background is in computer science as a computer engineer. He used this knowledge to help us engineer the best design.

Expert #2: Describe credential(s) and what makes them qualified as an expert.

Mr. Steinhorn is an electrical engineer who is an expert in constructing space equipment. He is qualified to help us find a creative solution.

Verification or Corrective Response to Design Review (E5)

Interview with Expert #1 informs corrective response:

Sid recommended that we continue to use the foam as a material to help store the items. We went with his recommendation because the foam boards are effective at storing items by cushioning them.

Interview with Expert #2 informs corrective response:

Mr. Steinhorn told us to reevaluate our measurements because it might not fit the console. We remeasured our console space and found a small error in our previous measurements (The width was too large) and fixed it thanks to his recommendation.